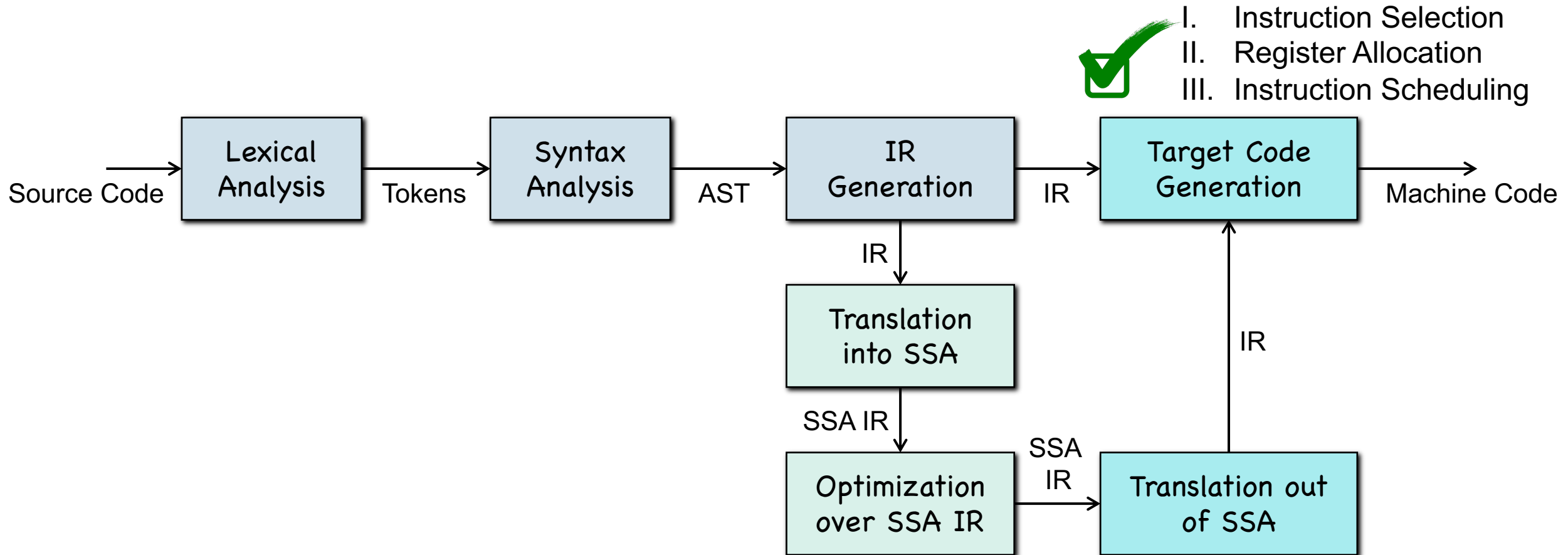


Recap-3

The Compilers' Back End

Back End of Compilers



PART I: Instruction Selection

Instructions

- Three categories of instructions
 - Load/Store
 - LD $R0, addr$
 - LD $R0, R1$
 - LD $R0, \#500$
 - ST $addr, R0$
 - Calculation
 - OP dst, src_1, src_2
 - *e.g.*, SUB $R0, R1, R2$
 - Jump
 - BR $L/addr$
 - Bcond $R, L/addr$
 - *e.g.*, BLTZ R, L

Addressing Modes

- What are addressing modes?

- Addressing Modes

- LD R1, a(R2)
- LD R1, 100(R2)
- LD R1, *R2
- LD R1, *100(R2)
- LD R1, #100

Addressing Modes: How we compute the addr

- LD R, *addr*
- ST *addr*, R

Note

假设 a 是一个元素为 8 字节值(比如实数)的数组。再假设 a 的元素的下标从 0 开始。我们
可以通过下面的指令序列来执行三地址指令 $b = a[i]$:

```
LD  R1, i           // R1 = i
MUL R1, R1, 8        // R1 = R1 * 8
LD  R2, a(R1)        // R2 = contents(a + contents(R1))
ST  b, R2            // b = R2
```

这里的第二步计算 $8i$; 而第三步把 a 的第 i 个元素的值放到 $R2$ 中, 这个元素位于离数组 a 的基地址 $8i$ 个字节的地方。

类似地, 三地址指令 $a[j] = c$ 所代表的对数组 a 的赋值可以实现为:

```
LD  R1, c           // R1 = c
LD  R2, j           // R2 = j
MUL R2, R2, 8        // R2 = R2 * 8
ST  a(R2), R1        // contents(a + contents(R2)) = R1
```

PART II: Register Allocation

Big Picture

- Why register allocation?
- Local register allocation
- Global register allocation

Spilling

- What is spilling?

Local Register Allocation

- Steps of local register allocation
- 1. Compute MAXLIVE
- 2. if $\text{MAXLIVE} \leq k$, ...
- 3. if $\text{MAXLIVE} > k$, ...

Local Register Allocation

1. LD	R1	#1028	// R1
2. LD	R2	*R1	// R1 R2
3. MUL	R3	R1 R2	// R1 R2 R3
4. LD	R4	x	// R1 R2 R3 R4
5. SUB	R5	R4 R2	// R1 R3 R5
6. LD	R6	z	// R1 R3 R5 R6
7. MUL	R7	R5 R6	// R1 R3 R7
8. SUB	R8	R7 R3	// R1 R8
9. ST	*R1	R8	//

MAXLIVE = 4

if $k \geq 4$, e.g., $k = 4$

Local Register Allocation

1. LD	R1	#1028	// R1
2. LD	R2	*R1	// R1 R2
3. MUL	R3	R1 R2	// R1 R2 R3
4. LD	R4	x	// R1 R2 R3 R4
5. SUB	R5	R4 R2	// R1 R3 R5
6. LD	R6	z	// R1 R3 R5 R6
7. MUL	R7	R5 R6	// R1 R3 R7
8. SUB	R8	R7 R3	// R1 R8
9. ST	*R1	R8	//

MAXLIVE = 4

if $k \geq 4$, e.g., $k = 4$

Local Register Allocation

1. LD	R1	#1028		// R1	
2. LD	R2	*R1		// R1 R2	
3. MUL	R3	R1	R2	// R1 R2 R3	
4. LD	R4	x		// R1 R2 R3 R4	
5. SUB	R2	R4	R2	// R1 R3 R2	
6. LD	R4	z		// R1 R3 R2 R4	
7. MUL	R2	R2	R4	// R1 R3 R2	
8. SUB	R2	R2	R3	// R1	R2
9. ST	*R1	R2		//	

MAXLIVE = 4

if $k \geq 4$, e.g., $k = 4$

Local Register Allocation

1. LD	R1	#1028		// R1	
2. LD	R2	*R1		// R1 R2	
3. MUL	R3	R1	R2	// R1 R2 R3	Spill R3 by <u>ST v R3</u>
4. LD	R4	x		// R1 R2 R3 R4	
5. SUB	R5	R4	R2	// R1 R3 R5	
6. LD	R6	z		// R1 R3 R5 R6	
7. MUL	R7	R5	R6	// R1 R3 R7	
8. SUB	R8	R7	R3	// R1 R8	
9. ST	*R1	R8		//	

MAXLIVE = 4

if $k < 4$, e.g., 3

Local Register Allocation

1. LD	R1	#1028		// R1	
2. LD	R2	*R1		// R1 R2	
3. MUL	R3	R1	R2	// R1 R2 R3	
4. LD	R4	x		// R1 R2 R3	R4
5. SUB	R5	R4	R2	// R1	R3 R5
6. LD	R6	z		// R1	R3 R5 R6
7. MUL	R7	R5	R6	// R1	R3 R7
8. SUB	R8	R7	R3	// R1	R8
9. ST	*R1	R8		//	

Spill R3 by ST v R3

MAXLIVE = 4

if $k < 4$, e.g., 3

Local Register Allocation

1. LD	R1	#1028		// R1	
2. LD	R2	*R1		// R1 R2	
3. MUL	R3	R1	R2	// R1 R2 R3	Spill R3 by <u>ST v R3</u>
4. LD	R4	x		// R1 R2	R4
5. SUB	R5	R4	R2	// R1	R5
6. LD	R6	z		// R1	R5 R6
7. MUL	R7	R5	R6	// R1	R7
8. SUB	R8	R7	R3	// R1	R8
9. ST	*R1	R8		//	

MAXLIVE = 4
if $k < 4$, e.g., 3

Local Register Allocation

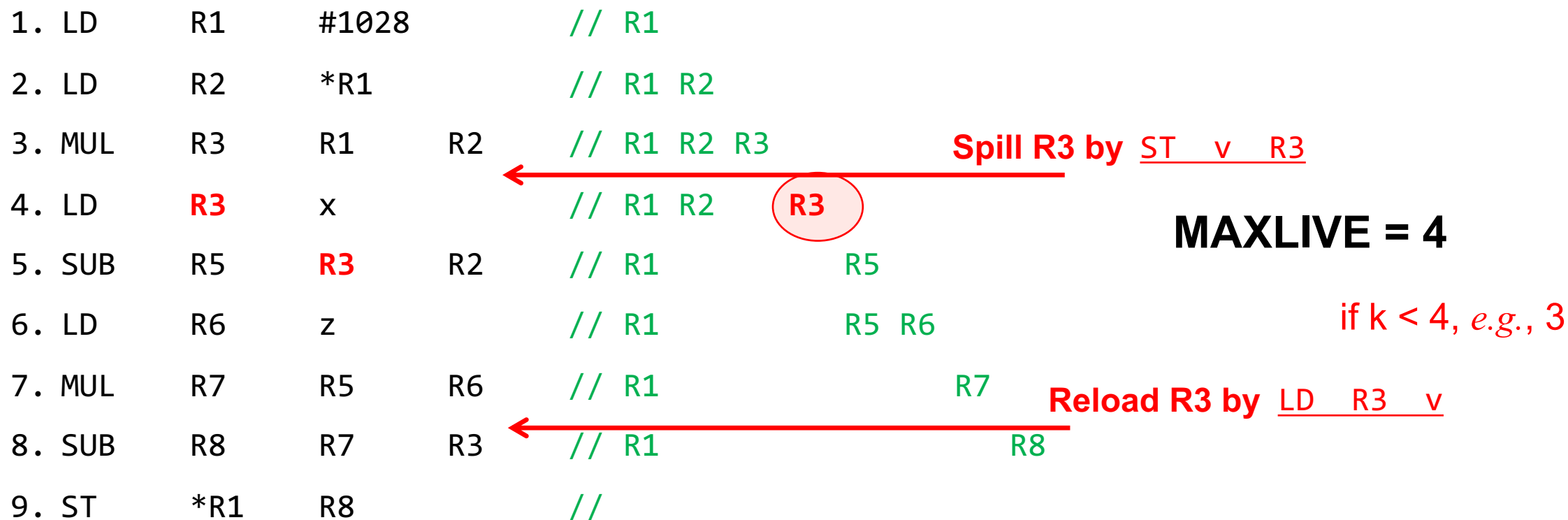
1. LD	R1	#1028	// R1	
2. LD	R2	*R1	// R1 R2	
3. MUL	R3	R1 R2	// R1 R2 R3	Spill R3 by <u>ST v R3</u>
4. LD	R4	x	// R1 R2	R4
5. SUB	R5	R4 R2	// R1	R5
6. LD	R6	z	// R1	R5 R6
7. MUL	R7	R5 R6	// R1	R7
8. SUB	R8	R7 R3	// R1	R8
9. ST	*R1	R8	//	

MAXLIVE = 4

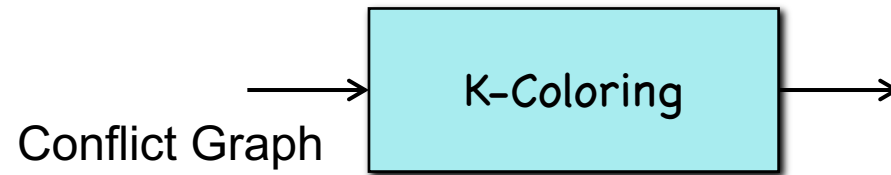
if $k < 4$, e.g., 3

Reload R3 by LD R3 v

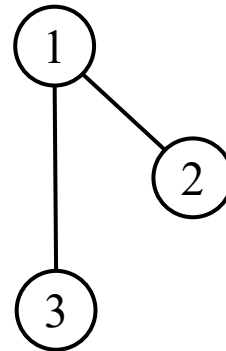
Local Register Allocation



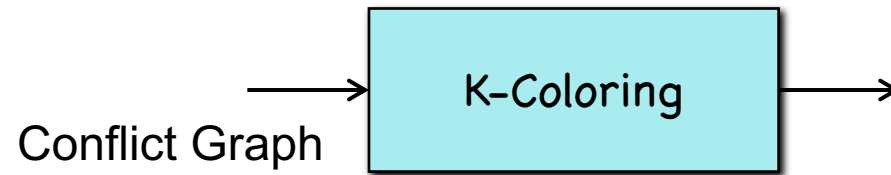
Global Register Allocation



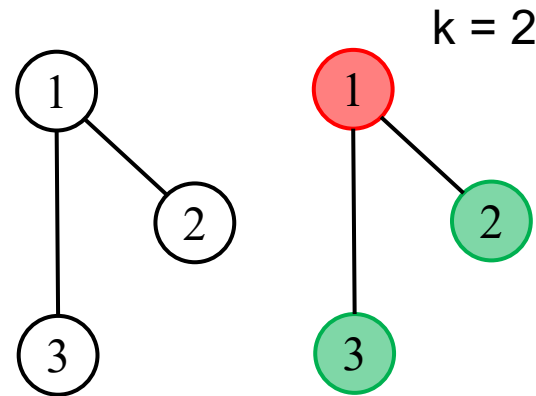
1.	LD	R1	#1028		// R1
2.	LD	R2	*R1		// R1 R2
3.	ST	a	R2		// R1
4.	MUL	R3	R1	R1	// R1 R3
5.	ST	x	R3		// R1
6.	LD	R2	a		// R1 R2
7.	ST	y	R2		// R1
8.	ST	z	R1		//



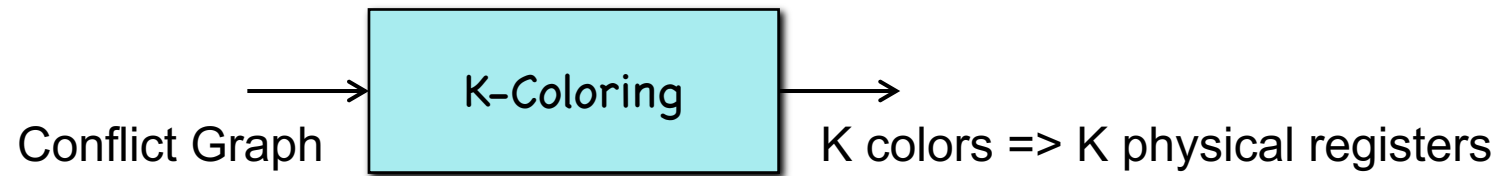
Global Register Allocation



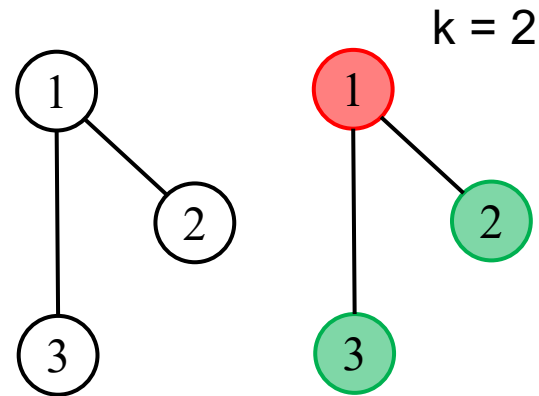
1.	LD	R1	#1028		// R1	
2.	LD	R2	*R1		// R1 R2	
3.	ST	a	R2		// R1	
4.	MUL	R3	R1	R1	// R1	R3
5.	ST	x	R3		// R1	
6.	LD	R2	a		// R1 R2	
7.	ST	y	R2		// R1	
8.	ST	z	R1		//	



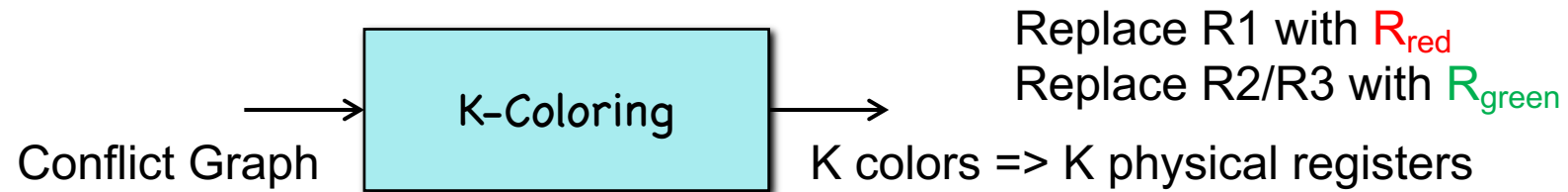
Global Register Allocation



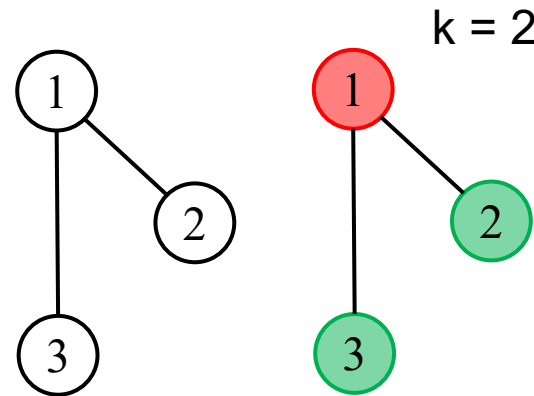
1.	LD	R1	#1028	// R1	
2.	LD	R2	*R1	// R1 R2	
3.	ST	a	R2	// R1	
4.	MUL	R3	R1 R1	// R1	R3
5.	ST	x	R3	// R1	
6.	LD	R2	a	// R1 R2	
7.	ST	y	R2	// R1	
8.	ST	z	R1	//	



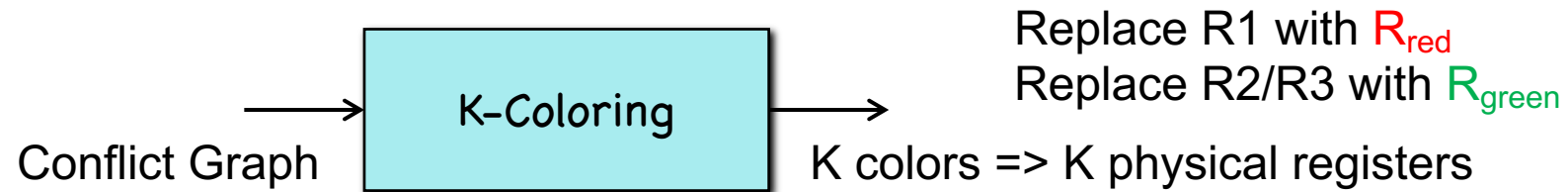
Global Register Allocation



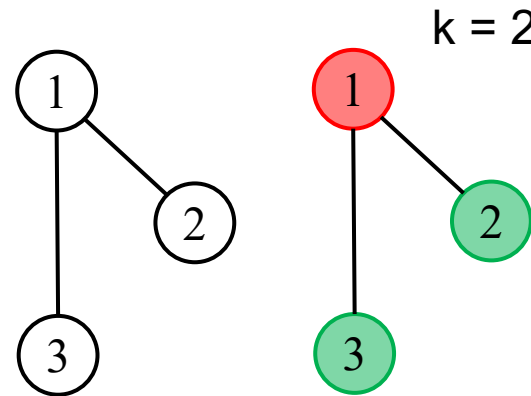
1.	LD	R1	#1028	//	R1	
2.	LD	R2	*R1	//	R1	R2
3.	ST	a	R2	//	R1	
4.	MUL	R3	R1 R1	//	R1	R3
5.	ST	x	R3	//	R1	
6.	LD	R2	a	//	R1	R2
7.	ST	y	R2	//	R1	
8.	ST	z	R1	//		



Global Register Allocation



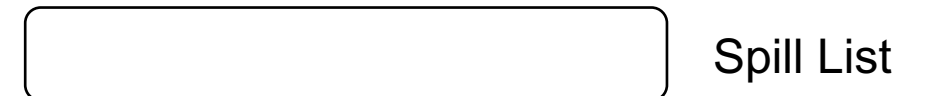
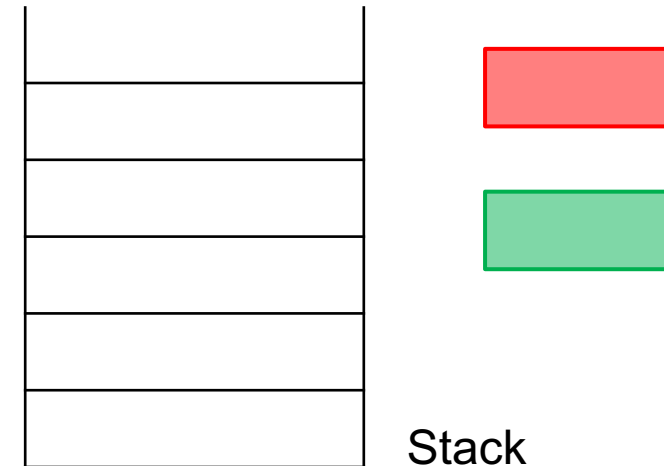
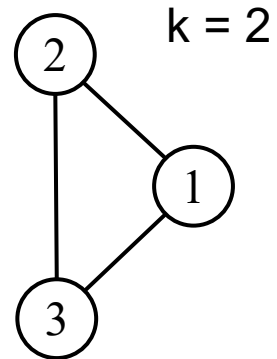
1.	LD	R1	#1028	//	R1	
2.	LD	R2	*R1	//	R1	R2
3.	ST	a	R2	//	R1	
4.	MUL	R3	R1	R1	//	R1 R3
5.	ST	x	R3	//	R1	
6.	LD	R2	a	//	R1	R2
7.	ST	y	R2	//	R1	
8.	ST	z	R1	//		



1.	LD	R_{red}	#1028	
2.	LD	R_{green}	* R_{red}	
3.	ST	a	R_{green}	
4.	MUL	R_{green}	R_{red}	R_{red}
5.	ST	x	R_{green}	
6.	LD	R_{green}	a	
7.	ST	y	R_{green}	
8.	ST	z	R_{red}	

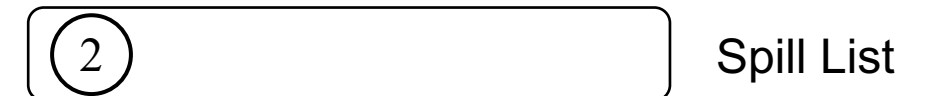
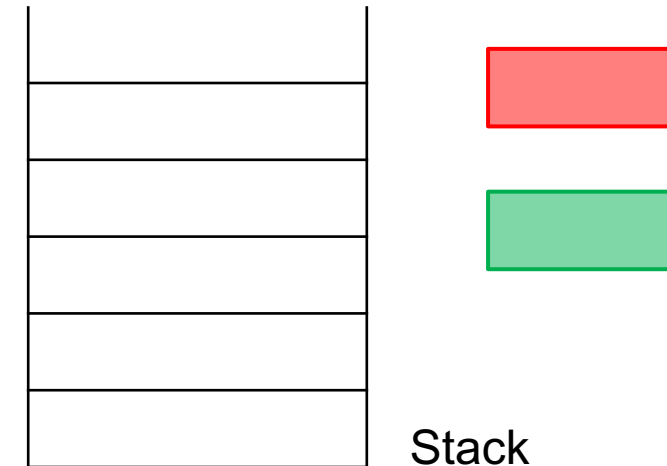
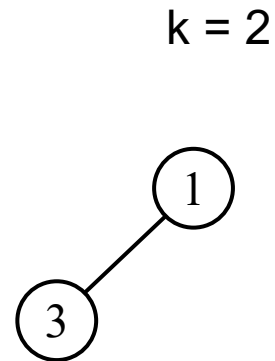
Chaitin's Algorithm

1.	LD	R1	#1028	// R1
2.	LD	R2	*R1	// R1 R2
3.	MUL	R3	R1 R2	// R1 R2 R3
4.	ST	x	R3	// R1 R2
5.	ST	y	R2	// R1
6.	ST	z	R1	//



Chaitin's Algorithm

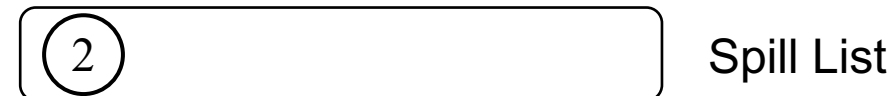
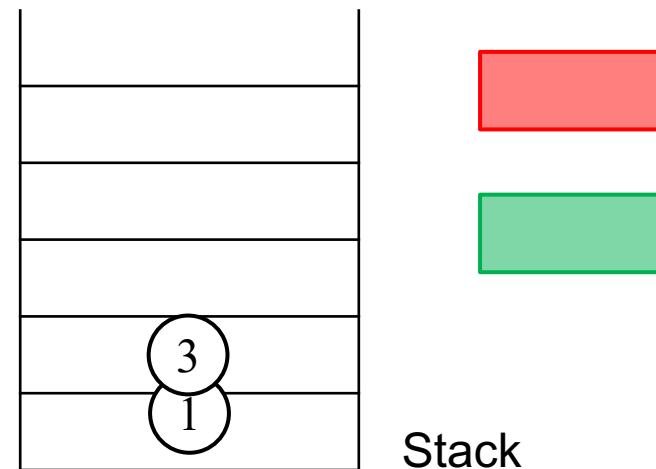
1.	LD	R1	#1028	// R1
2.	LD	R2	*R1	// R1 R2
3.	MUL	R3	R1 R2	// R1 R2 R3
4.	ST	x	R3	// R1 R2
5.	ST	y	R2	// R1
6.	ST	z	R1	//



Chaitin's Algorithm

1.	LD	R1	#1028	// R1
2.	LD	R2	*R1	// R1 R2
3.	MUL	R3	R1 R2	// R1 R2 R3
4.	ST	x	R3	// R1 R2
5.	ST	y	R2	// R1
6.	ST	z	R1	//

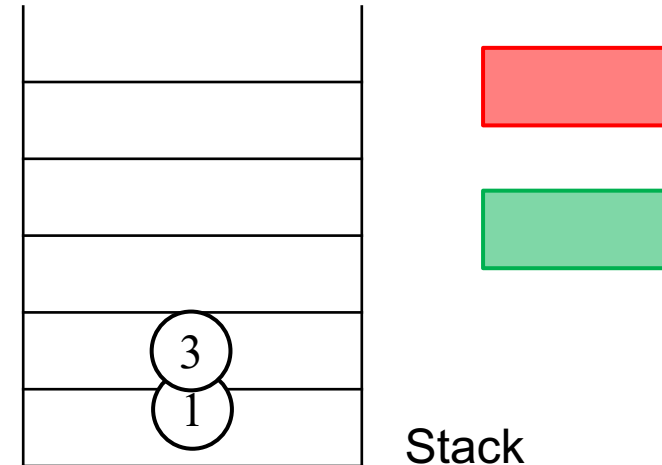
$k = 2$



Chaitin's Algorithm

1.	LD	R1	#1028	// R1
2.	LD	R2	*R1	// R1 R2
3.	MUL	R3	R1 R2	// R1 R2 R3
4.	ST	x	R3	// R1 R2
5.	ST	y	R2	// R1
6.	ST	z	R1	//

$k = 2$

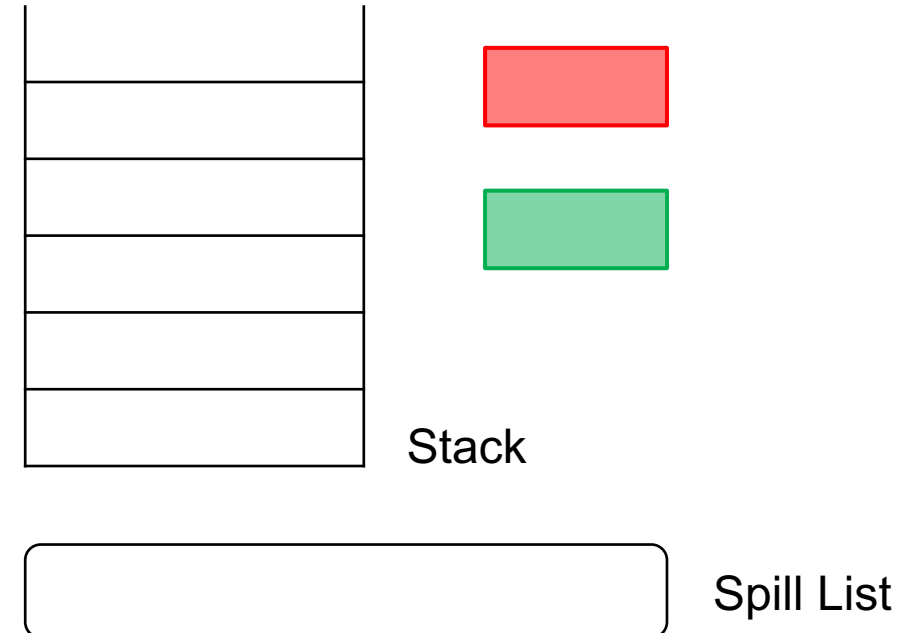


Spill R2!

Chaitin's Algorithm

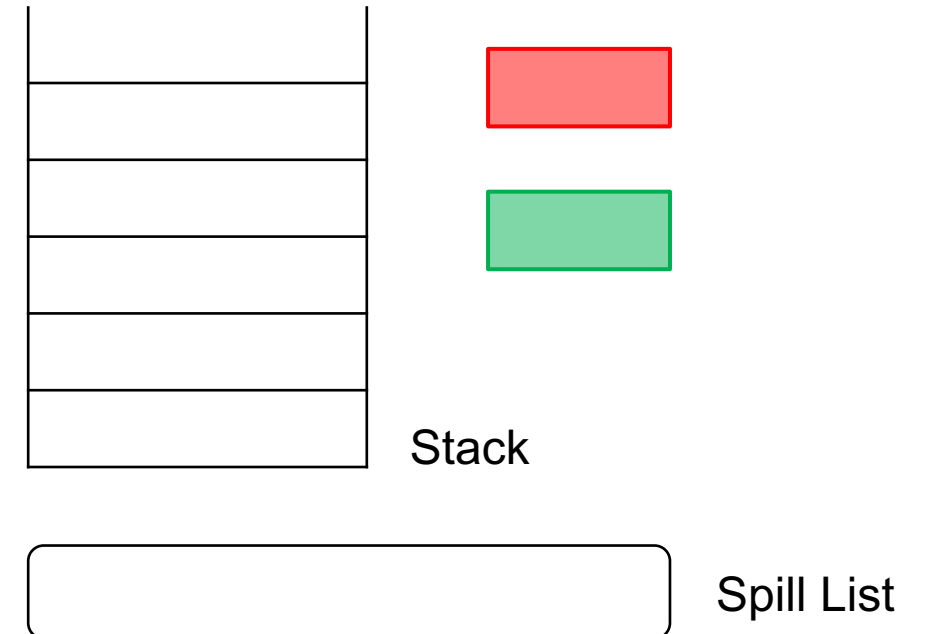
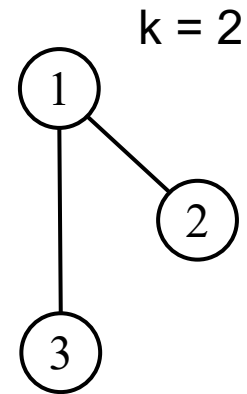
1.	LD	R1	#1028	//	R1	
2.	LD	R2	*R1	//	R1	R2
3.	ST	a	R2	//	R1	
4.	MUL	R3	R1 R1	//	R1	R3
5.	ST	x	R3	//	R1	
6.	LD	R2	a	//	R1	R2
7.	ST	y	R2	//	R1	
8.	ST	z	R1	//		

k = 2



Chaitin's Algorithm

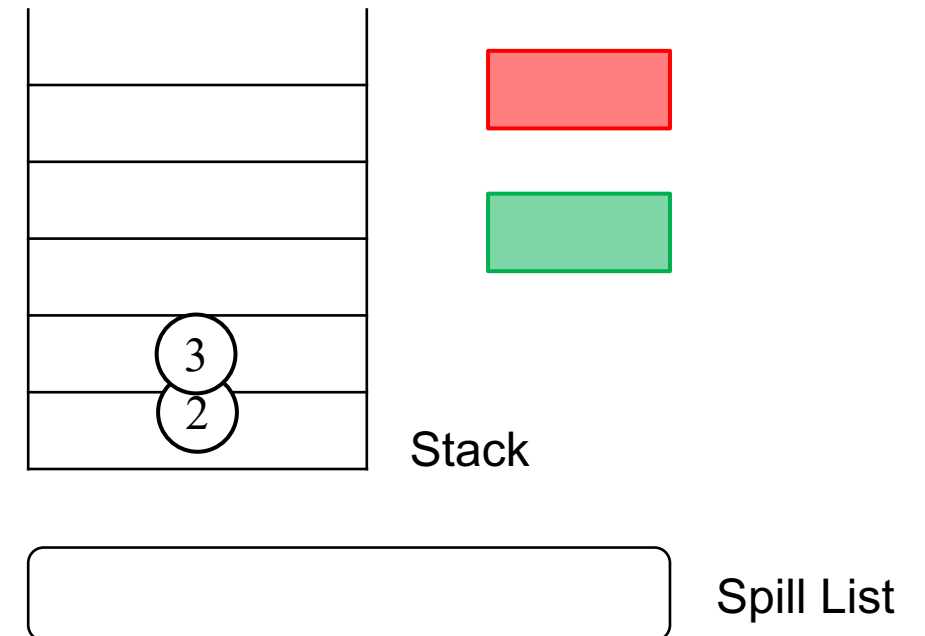
1.	LD	R1	#1028	// R1	
2.	LD	R2	*R1	// R1 R2	
3.	ST	a	R2	// R1	
4.	MUL	R3	R1 R1	// R1 R3	
5.	ST	x	R3	// R1	
6.	LD	R2	a	// R1 R2	
7.	ST	y	R2	// R1	
8.	ST	z	R1	//	



Chaitin's Algorithm

1.	LD	R1	#1028	// R1	
2.	LD	R2	*R1	// R1 R2	
3.	ST	a	R2	// R1	
4.	MUL	R3	R1 R1	// R1 R3	
5.	ST	x	R3	// R1	
6.	LD	R2	a	// R1 R2	
7.	ST	y	R2	// R1	
8.	ST	z	R1	//	

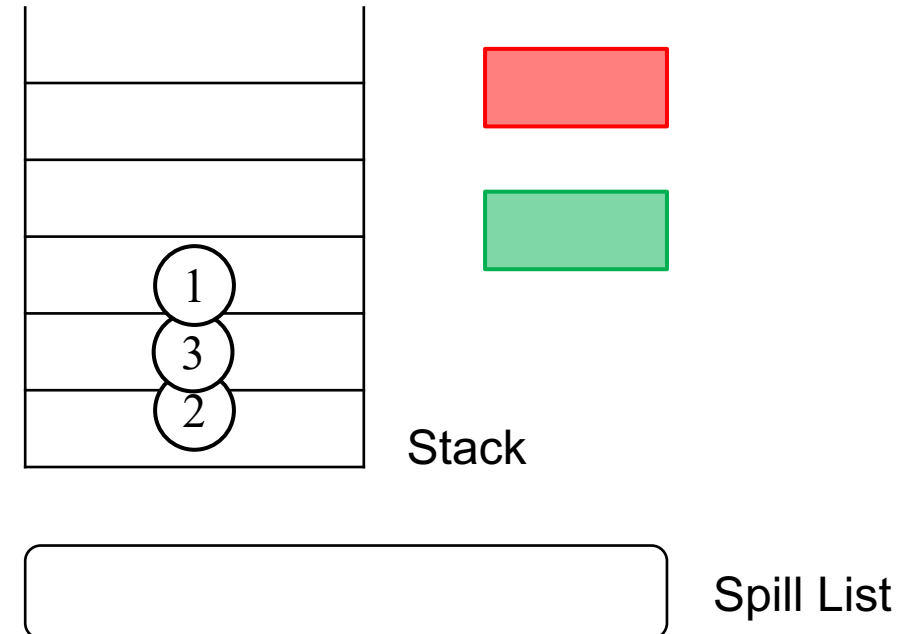
① $k = 2$



Chaitin's Algorithm

1.	LD	R1	#1028	//	R1	
2.	LD	R2	*R1	//	R1	R2
3.	ST	a	R2	//	R1	
4.	MUL	R3	R1 R1	//	R1	R3
5.	ST	x	R3	//	R1	
6.	LD	R2	a	//	R1	R2
7.	ST	y	R2	//	R1	
8.	ST	z	R1	//		

k = 2

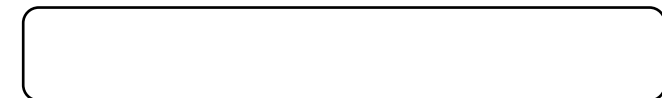
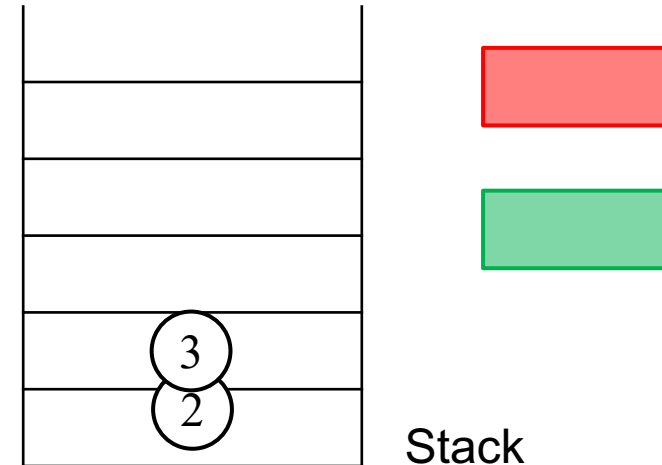


Chaitin's Algorithm

1.	LD	R1	#1028	// R1	
2.	LD	R2	*R1	// R1 R2	
3.	ST	a	R2	// R1	
4.	MUL	R3	R1 R1	// R1 R3	
5.	ST	x	R3	// R1	
6.	LD	R2	a	// R1 R2	
7.	ST	y	R2	// R1	
8.	ST	z	R1	//	

1

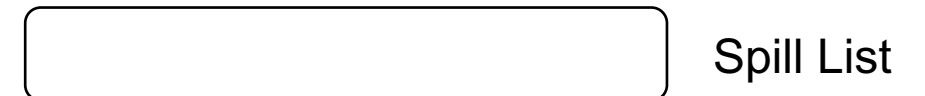
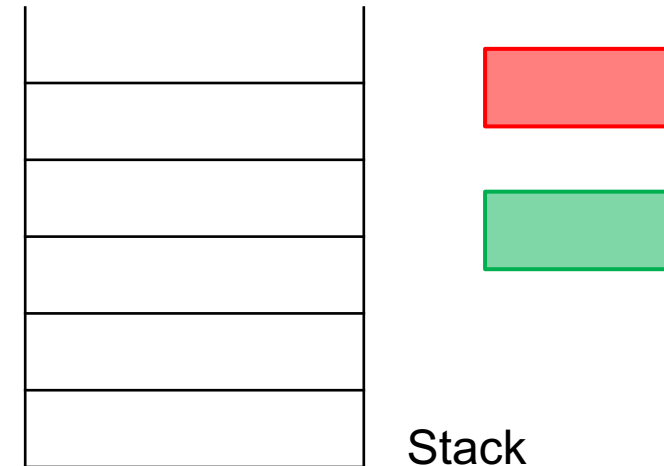
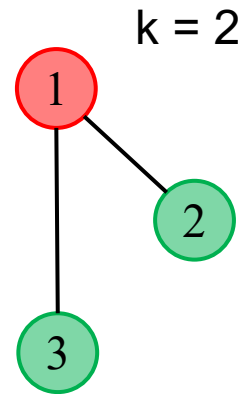
k = 2



Spill List

Chaitin's Algorithm

1.	LD	R1	#1028	// R1	
2.	LD	R2	*R1	// R1 R2	
3.	ST	a	R2	// R1	
4.	MUL	R3	R1 R1	// R1	R3
5.	ST	x	R3	// R1	
6.	LD	R2	a	// R1 R2	
7.	ST	y	R2	// R1	
8.	ST	z	R1	//	

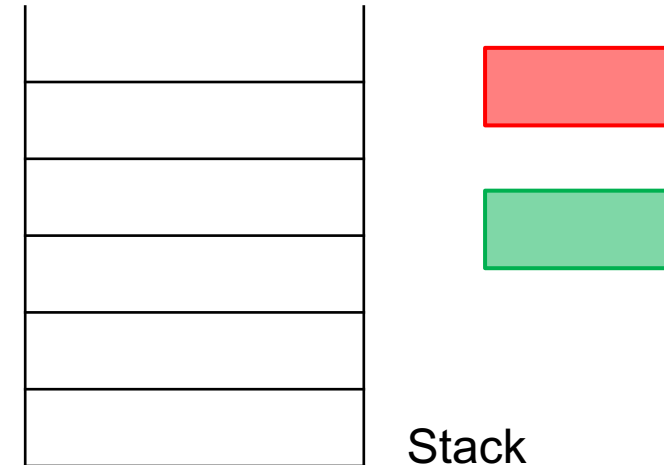
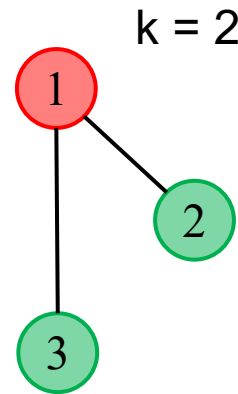


Chaitin's Algorithm

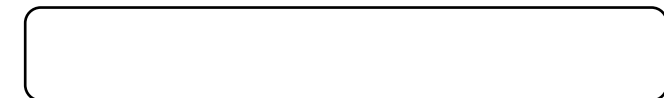
Replace R1 with R_{red}
Replace R2/R3 with R_{green}



1.	LD	R1	#1028	//	R1	
2.	LD	R2	*R1	//	R1	R2
3.	ST	a	R2	//	R1	
4.	MUL	R3	R1 R1	//	R1	R3
5.	ST	x	R3	//	R1	
6.	LD	R2	a	//	R1	R2
7.	ST	y	R2	//	R1	
8.	ST	z	R1	//		



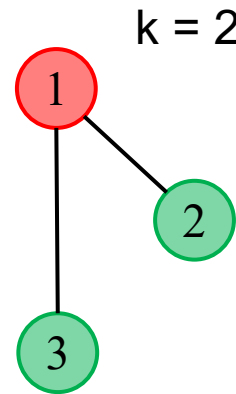
Stack



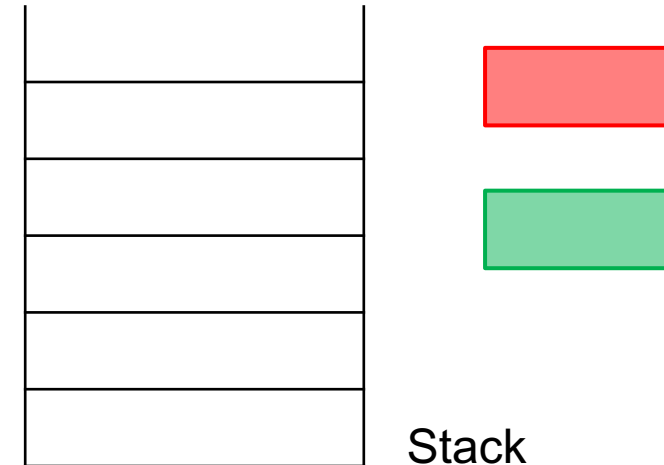
Spill List

Chaitin's Algorithm

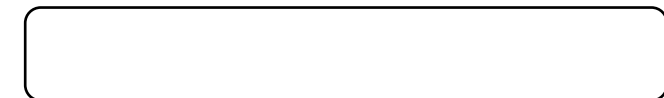
Replace R1 with R_{red}
Replace R2/R3 with R_{green}



1.	LD	R_{red}	#1028
2.	LD	R_{green}	* R_{red}
3.	ST	a	R_{green}
4.	MUL	R_{green}	R_{red} R_{red}
5.	ST	x	R_{green}
6.	LD	R_{green}	a
7.	ST	y	R_{green}
8.	ST	z	R_{red}



Stack



Spill List

PART III: Instruction Scheduling

Why Scheduling?

- Every modern high-performance processor can execute several operations in a single clock cycle.

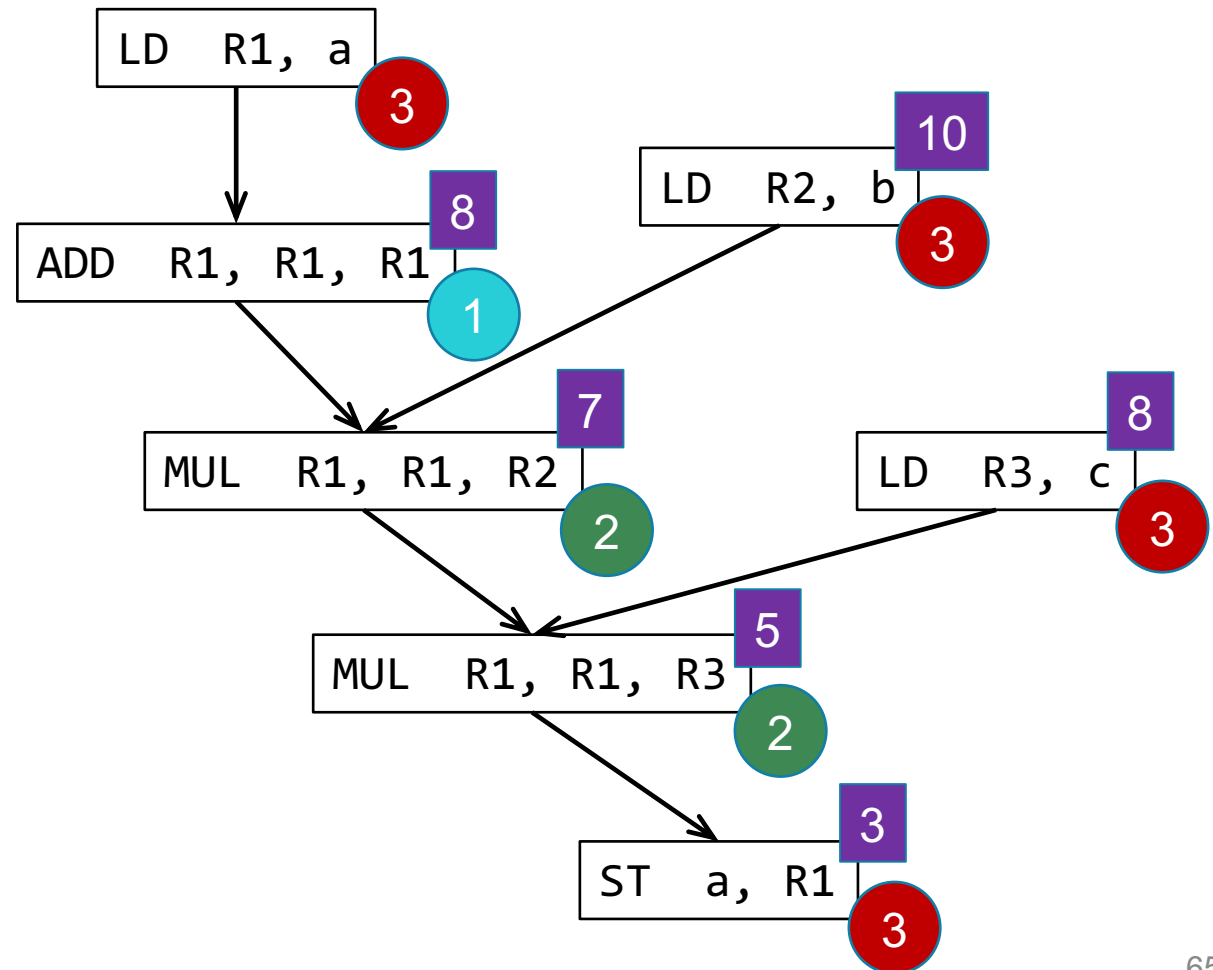
Why Compilers Matter?

- Hardware only can execute instructions that have been fetched
- Hardware has limited space to buffer stalled operations
- Compilers can place independent operations close together to allow better hardware utilization

In-Block Scheduling

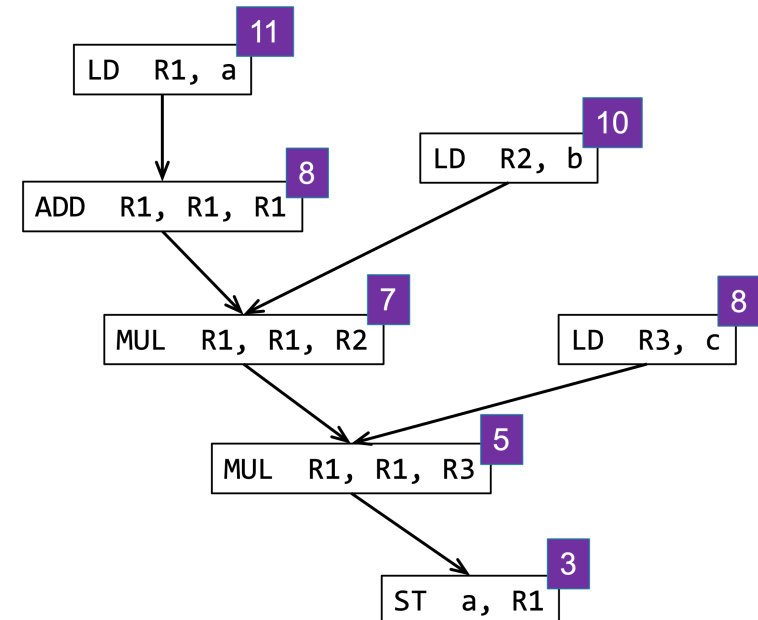
- True/Fake Dependence
- Dependence Graph
- List Scheduling

LD	R1,	a
ADD	R1,	R1, R1
LD	R2,	b
MUL	R1,	R1, R2
LD	R2,	c
MUL	R1,	R1, R2
ST	a,	R1



List Scheduling

- Check each clock cycle
- Schedule instructions when they are ready
- When multi instructions can be scheduled, check their **priority**
 - The longest latency path (max depth)
 - Using the most resources
 - ...



List Scheduling

- Cycle = 9, 10, 11

LD	R1,	a
LD	R2,	b
LD	R3,	c
ADD	R1,	R1, R1
MUL	R1,	R1, R2
NOP (no operation)		
MUL	R1,	R1, R3
NOP (no operation)		
ST	a,	R1

ST a, R1 3

PART IV: Exercises

Exercise - I

- Consider how we compile the following assignment to machine

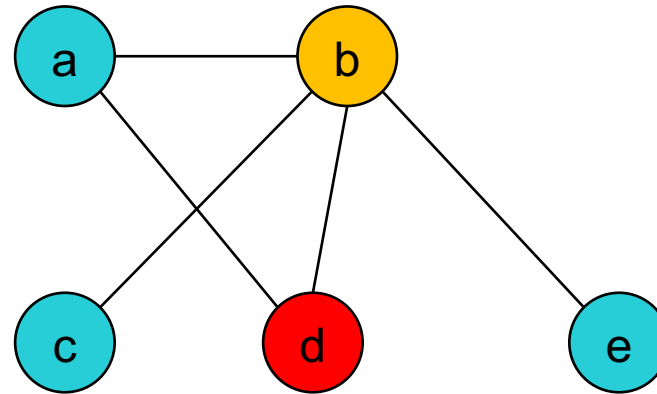
code: $e = (a - 1) * \left(\frac{(b+c)}{3} + d \right)$

- LD R, var; ST var, R
- OP R1, R2, R3, where OP can be ADD, SUB, MUL, DIV, ..., R2 and R3 can be immediate number
- **Q1: What's the purpose of register allocation? (4')**
- **Q2: What's the minimum registers required if spilling is not allowed (2')**
- **Q3: Write the machine code (4')**

Exercise - II

- Consider the following code, construct an interference graph

```
1  a = read();  
2  b = read();  
3  if (a > b) {  
4      c = read();  
5      e = c - b;  
6  } else {  
7      d = b;  
8      e = a + d;  
9  }  
10 print(b);  
11 print(e);
```



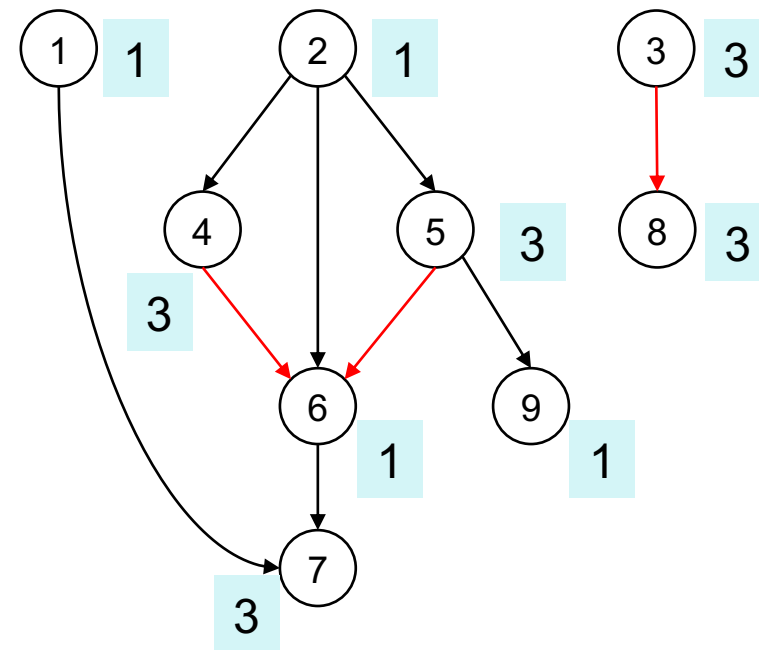
- How many colors are required to avoid spilling registers

Exercise - III

- Schedule the following code, assuming LD/ST needs 3 cycles and add needs 1 cycle (SP is a special register and cannot be renamed)

```

1.  ADD    R2,    R1,    #1
2.  ADD    SP,    12,    SP
3.  ST     a,     R0
4.  LD     R3,    -4(SP)
5.  LD     R4,    -8(SP)
6.  ADD    SP,    SP,    #8
7.  ST     0(SP), R2
8.  LD     R5,    a
9.  ADD    R4,    R4,    #1
    
```



Exercise - III

- Schedule the following code, assuming LD/ST needs 3 cycles and add needs 1 cycle (SP is a special register and cannot be renamed)

```

1.  ADD    R2,    R1,    #1
2.  ADD    SP,    12,    SP
3.  ST     a,     R0
4.  LD     R3,    -4(SP)
5.  LD     R4,    -8(SP)
6.  ADD    SP,    SP,    #8
7.  ST     0(SP), R2
8.  LD     R5,    a
9.  ADD    R4,    R4,    #1
  
```

